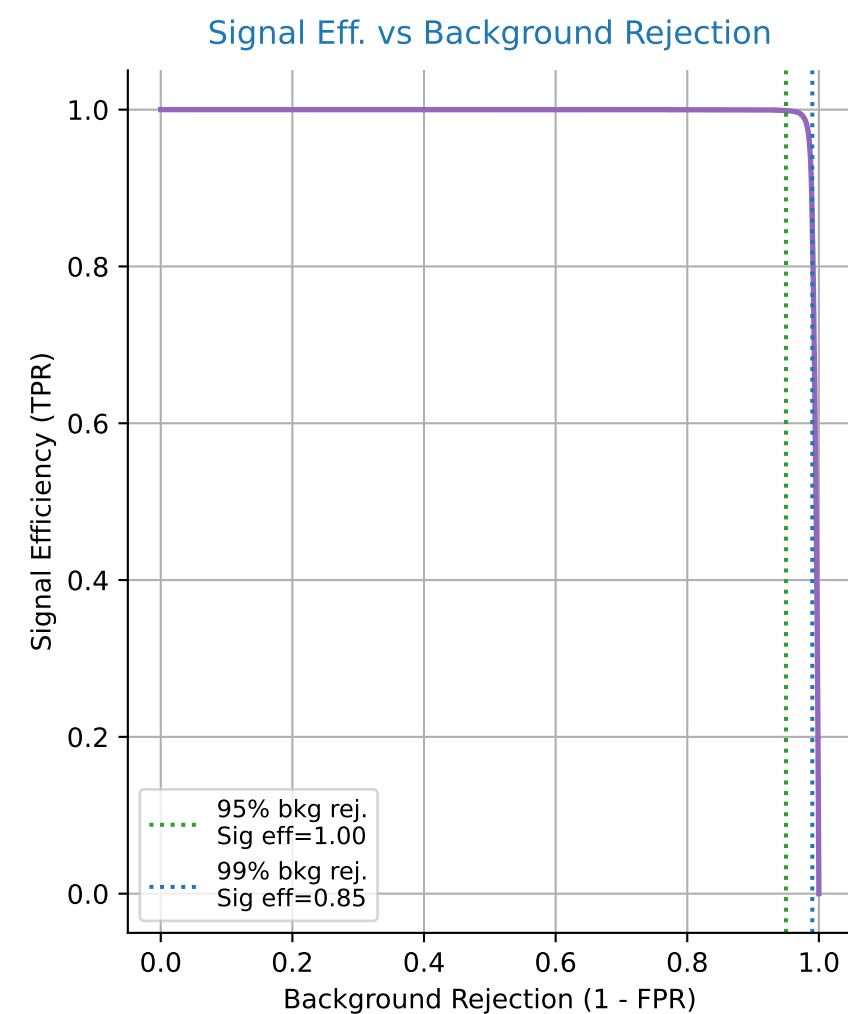
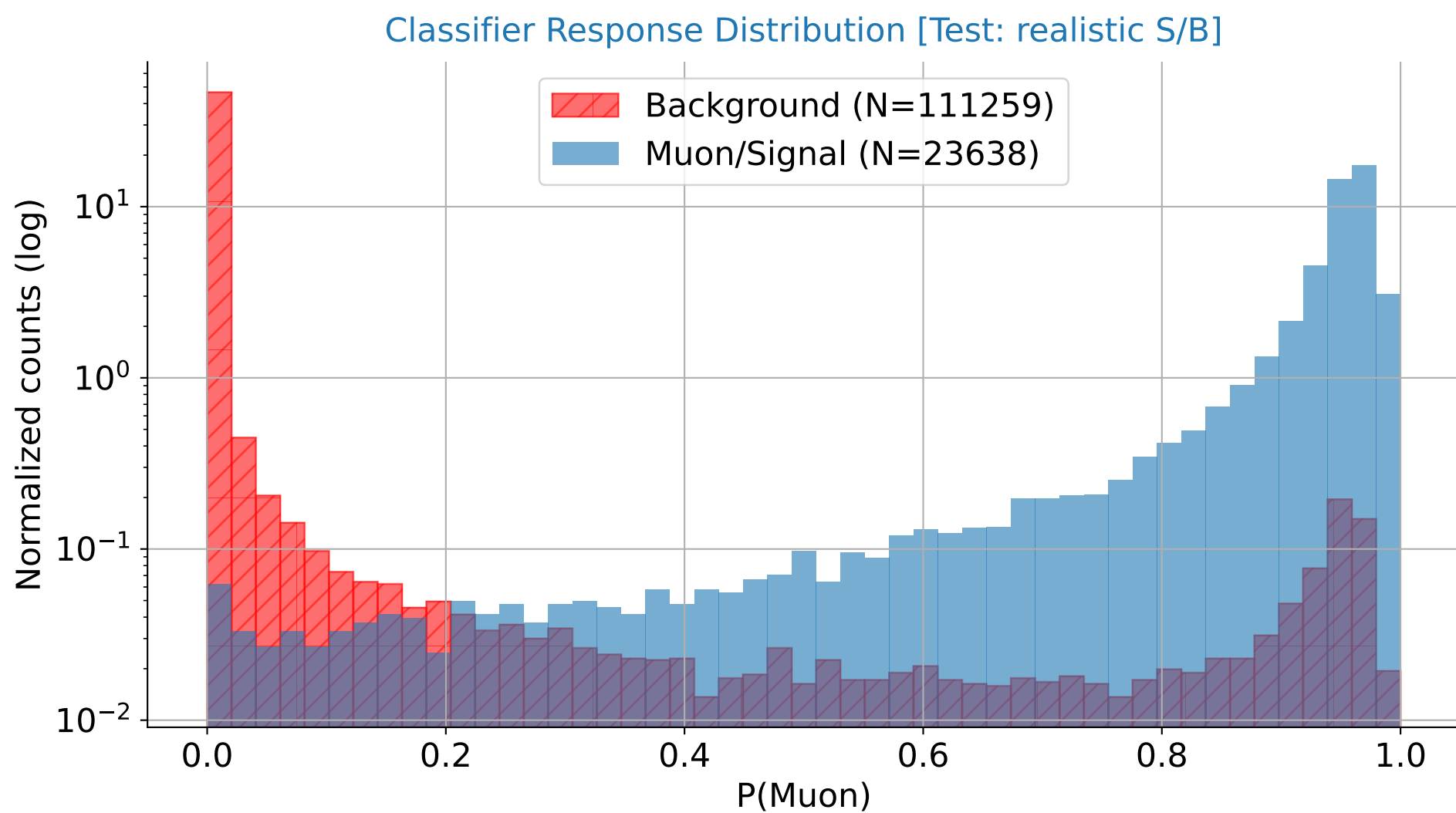
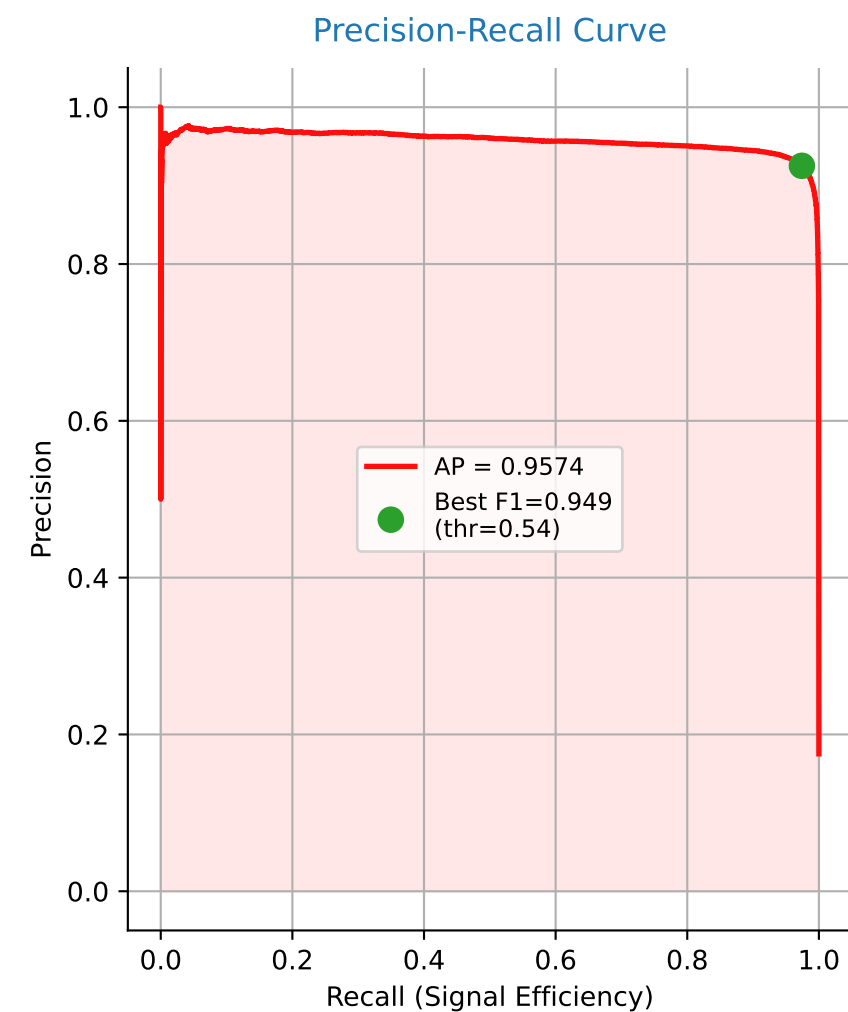
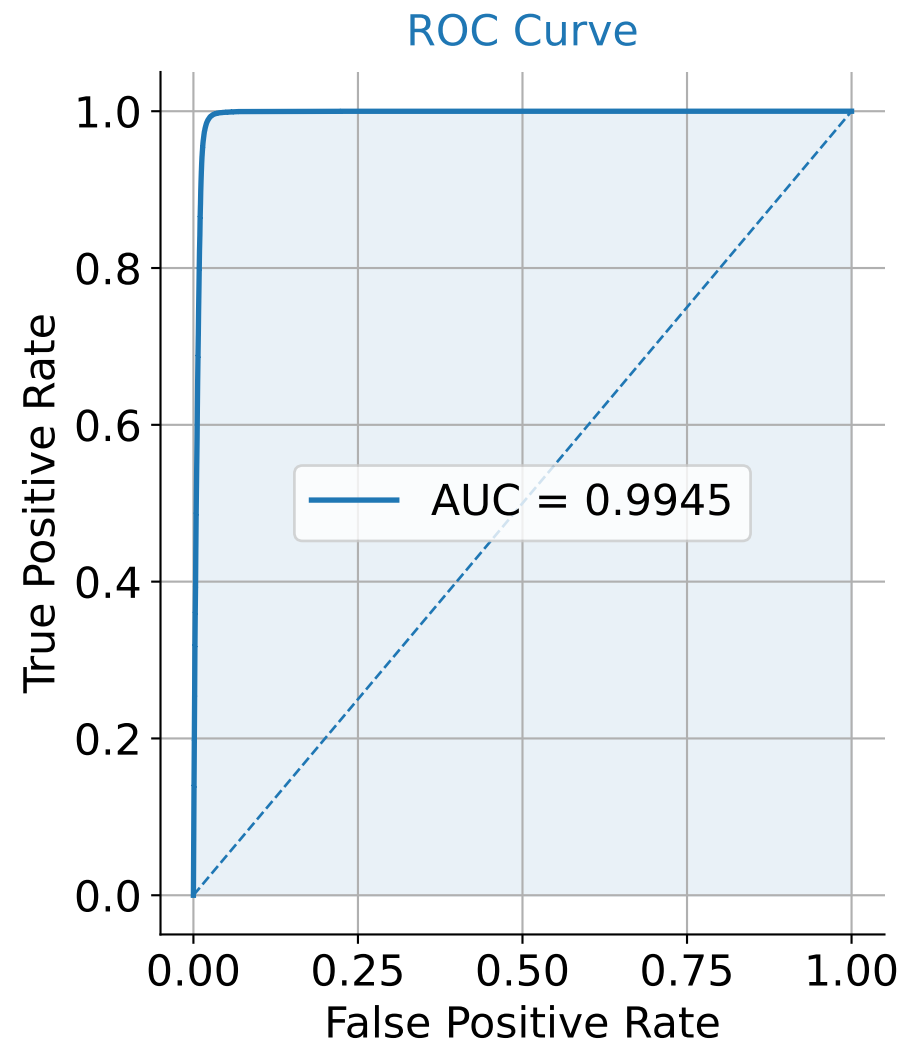
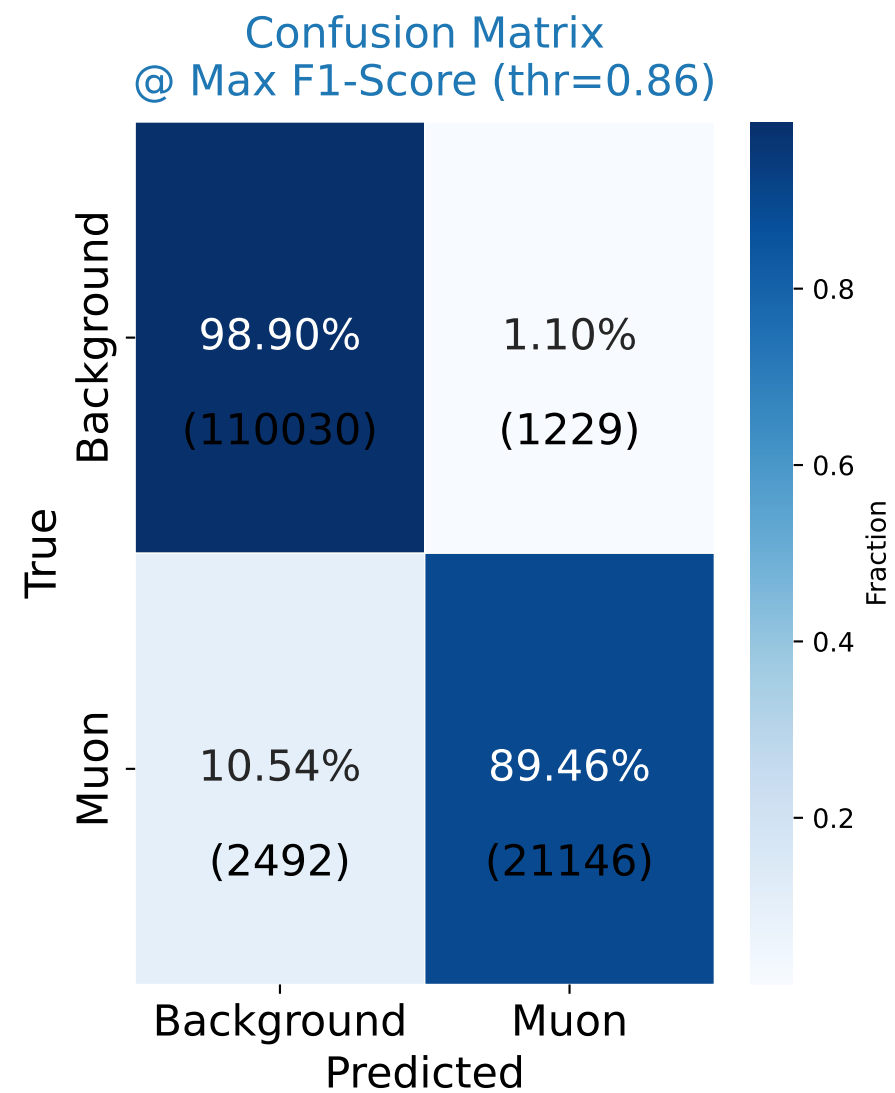


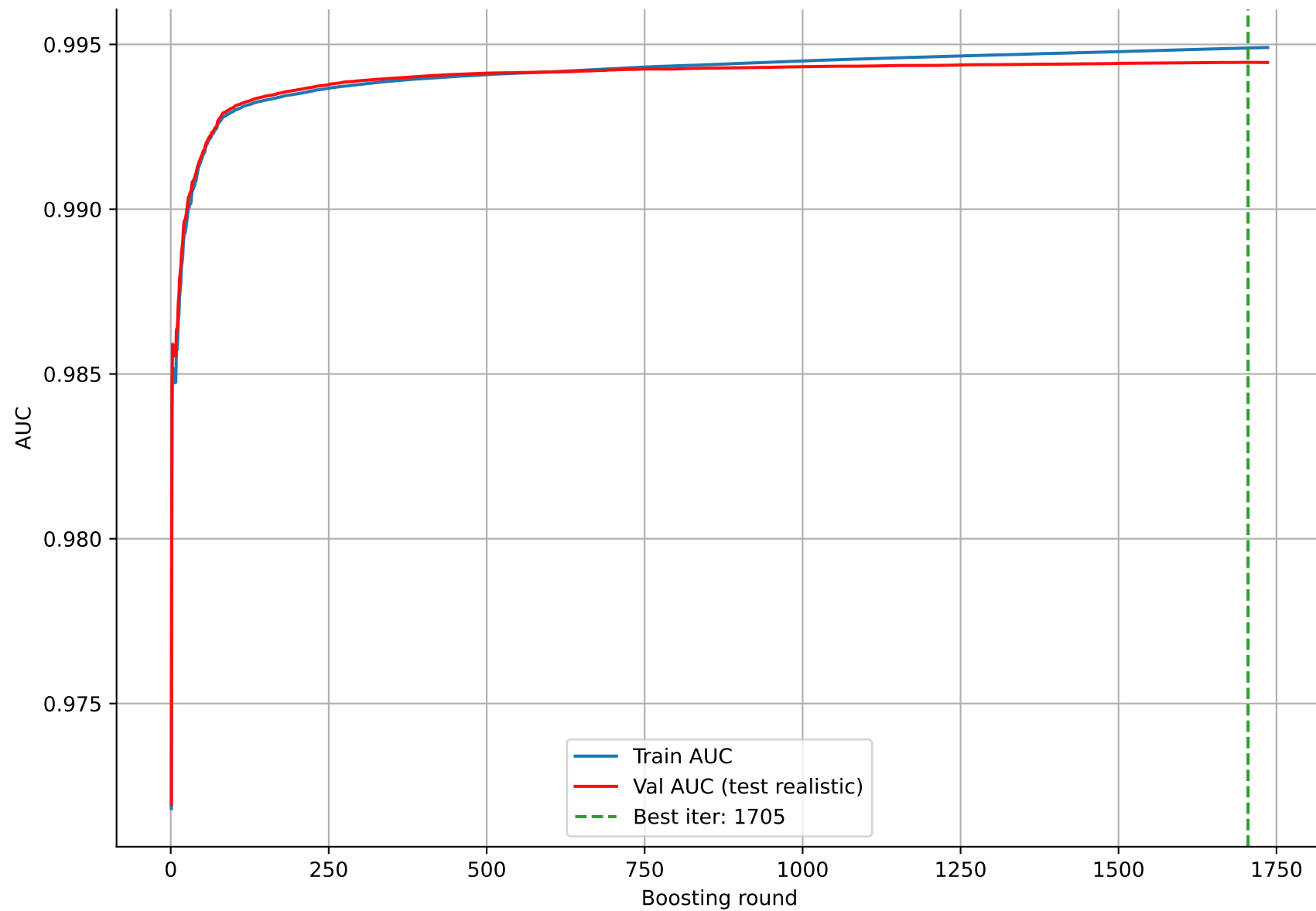
Muon Identification — XGBoost Analysis

[Train S/B=1:5 | Test S/B=1:5 (realistic, all files)]

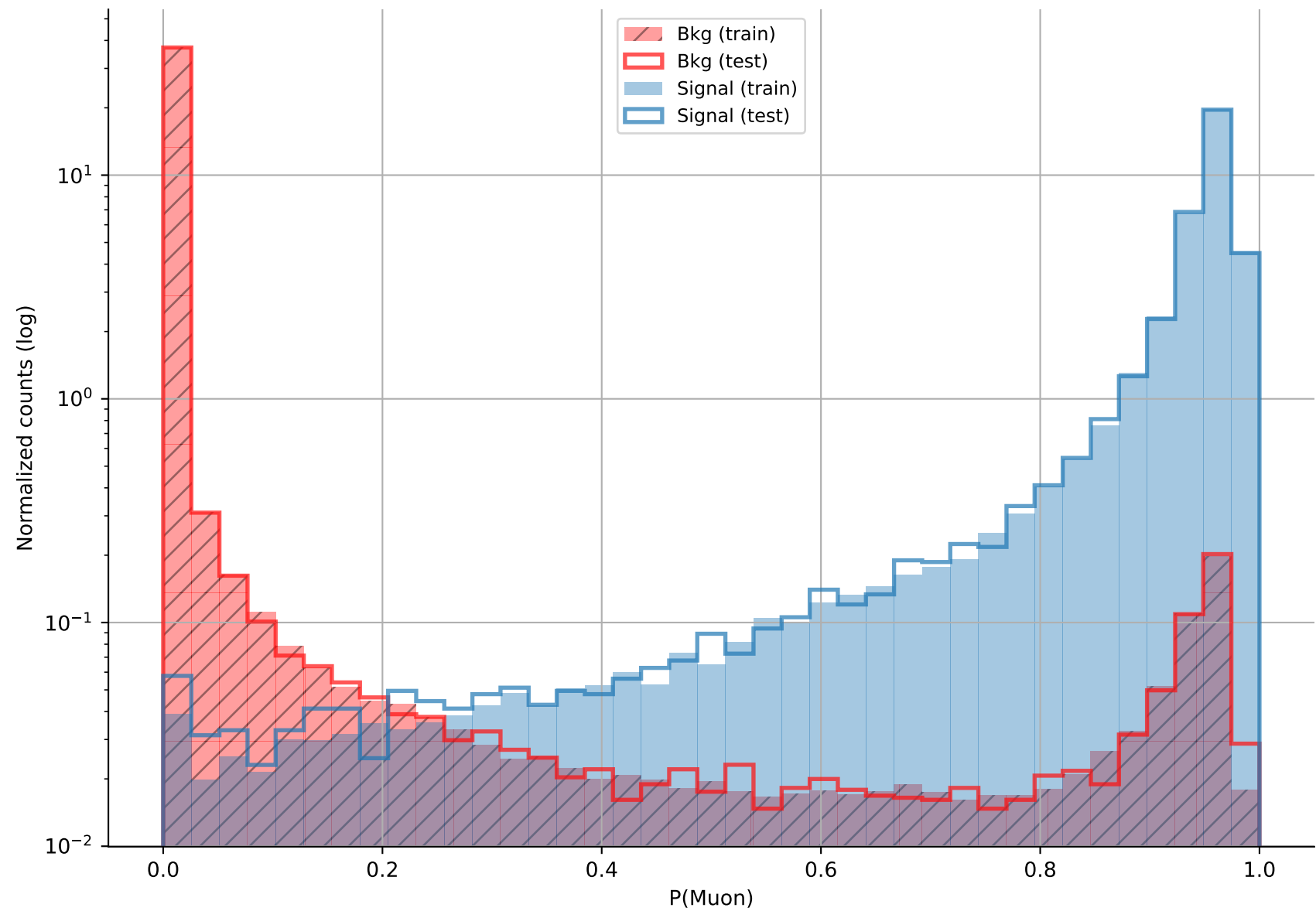


XGBoost Training Diagnostics

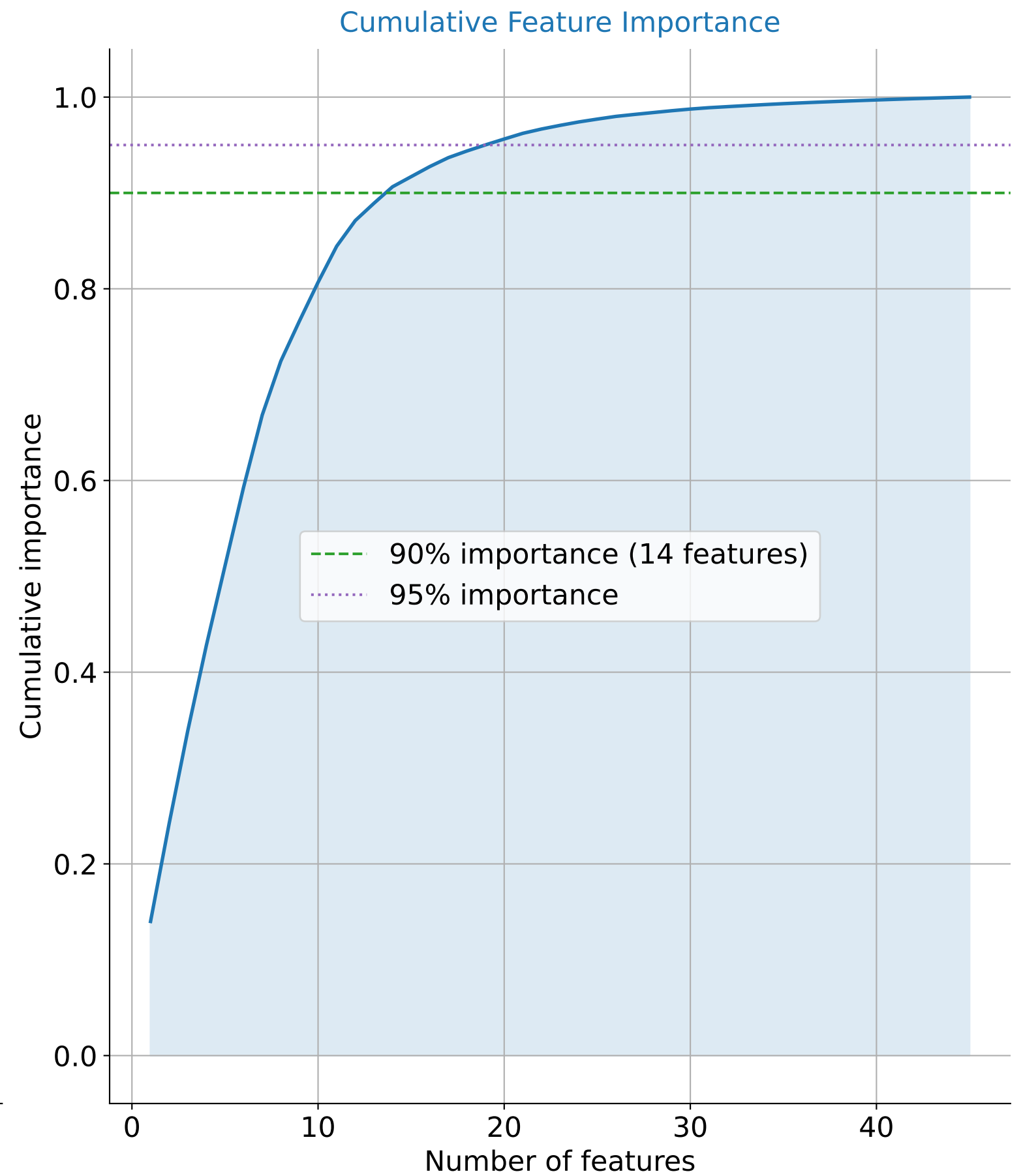
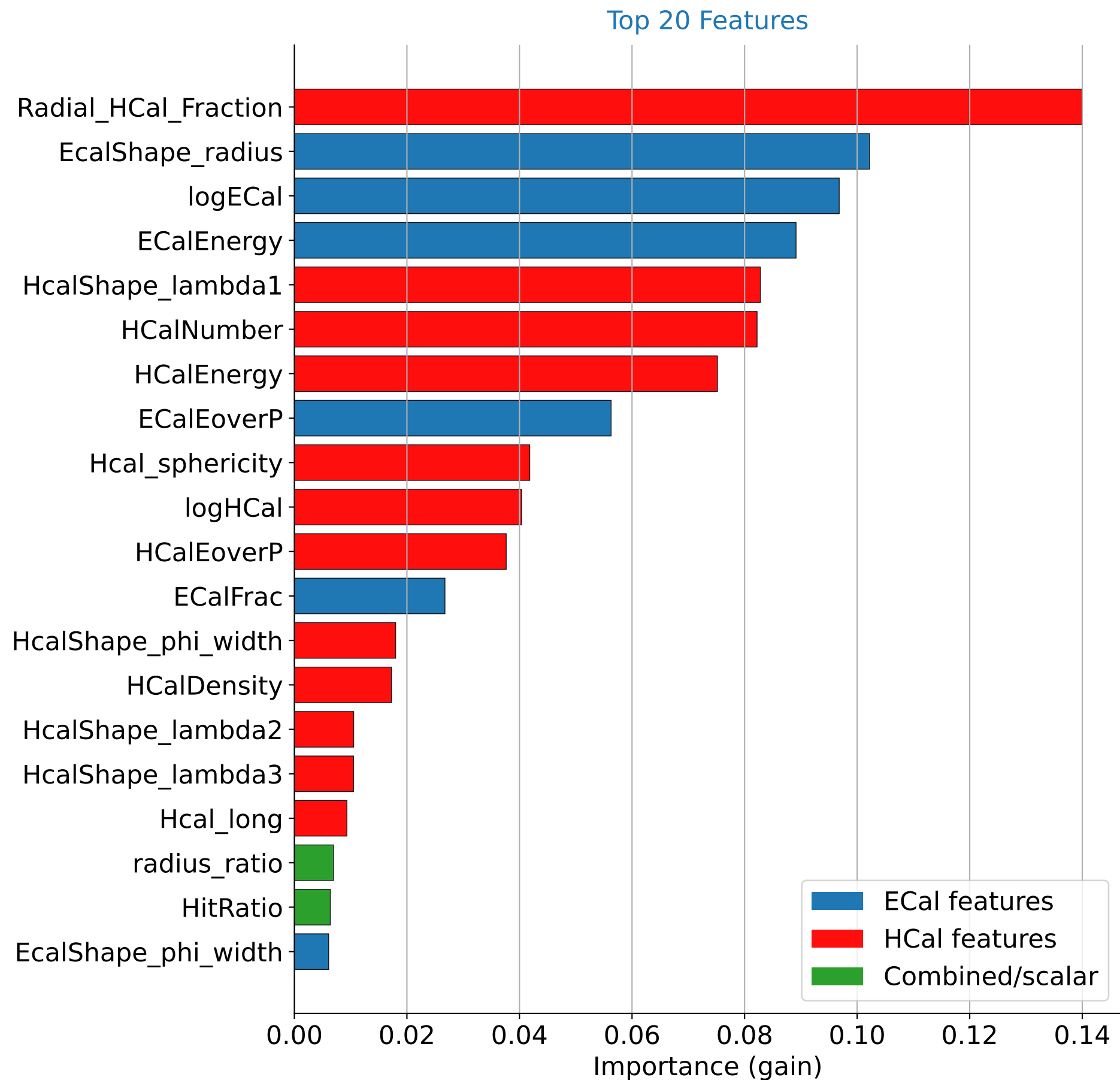
AUC vs Boosting Rounds



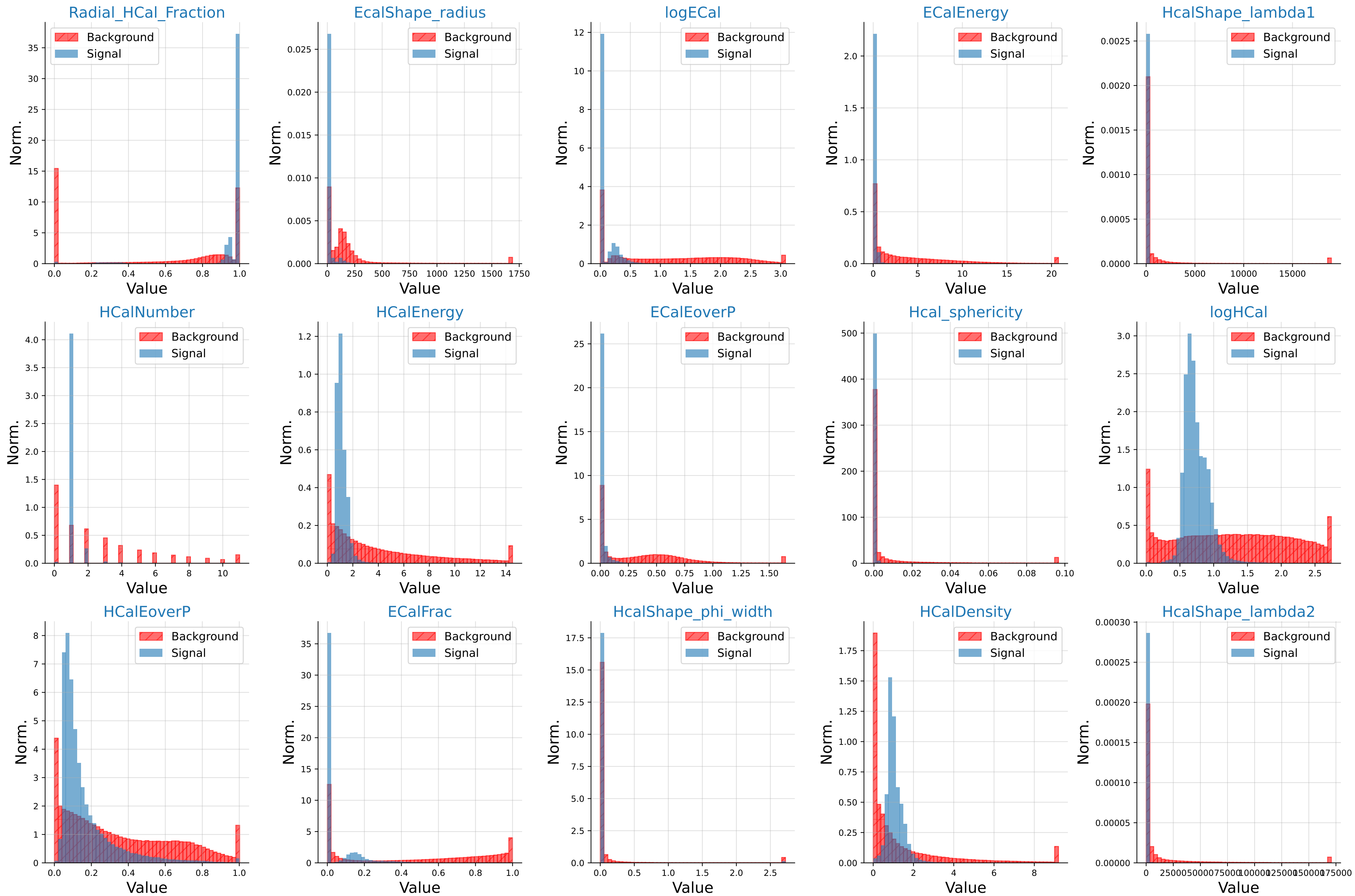
Overtraining Check (Train vs Test)



Feature Importances — XGBoost

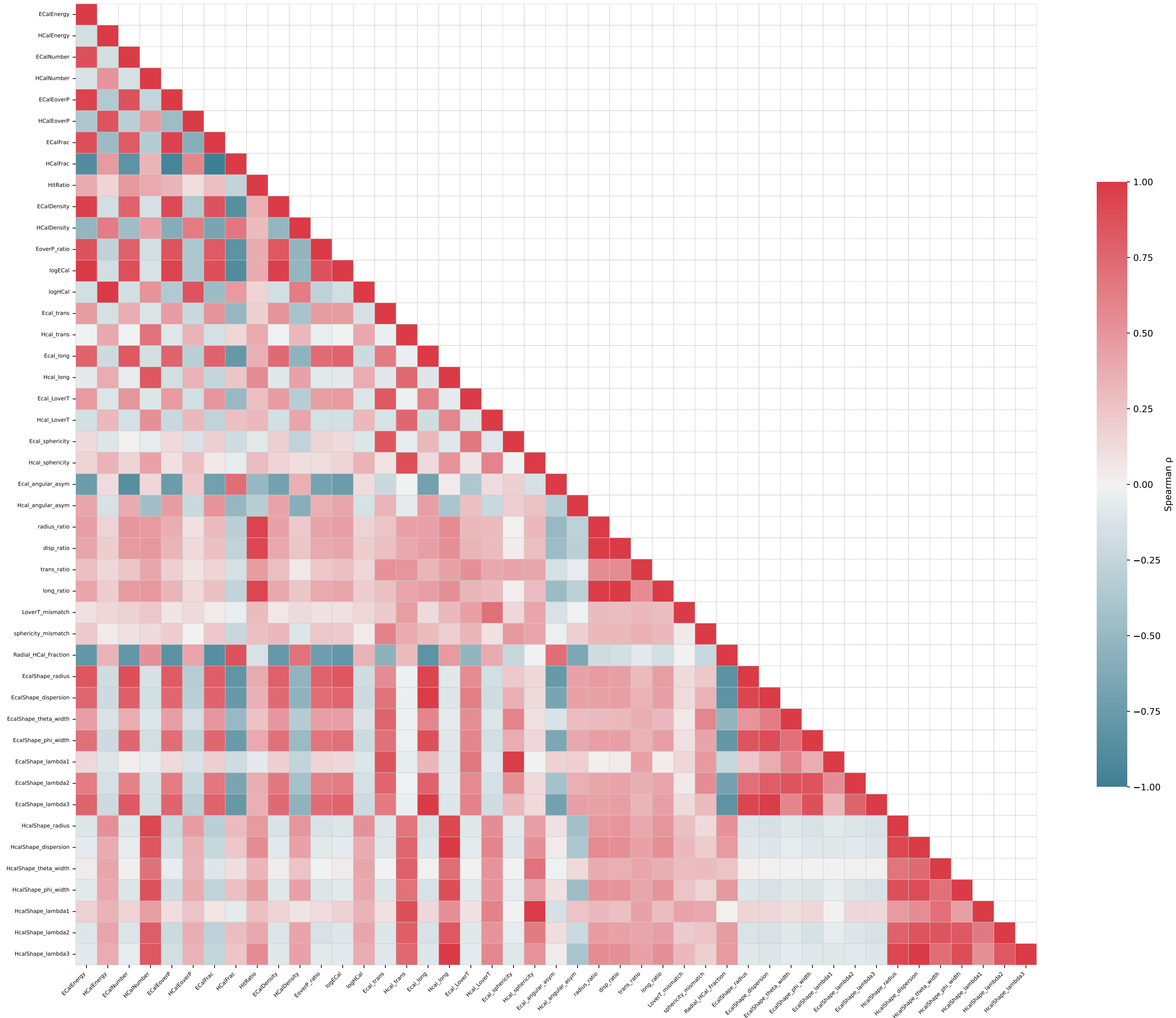


Key Physics Variable Distributions: Signal vs Background [full dataset before splitting]

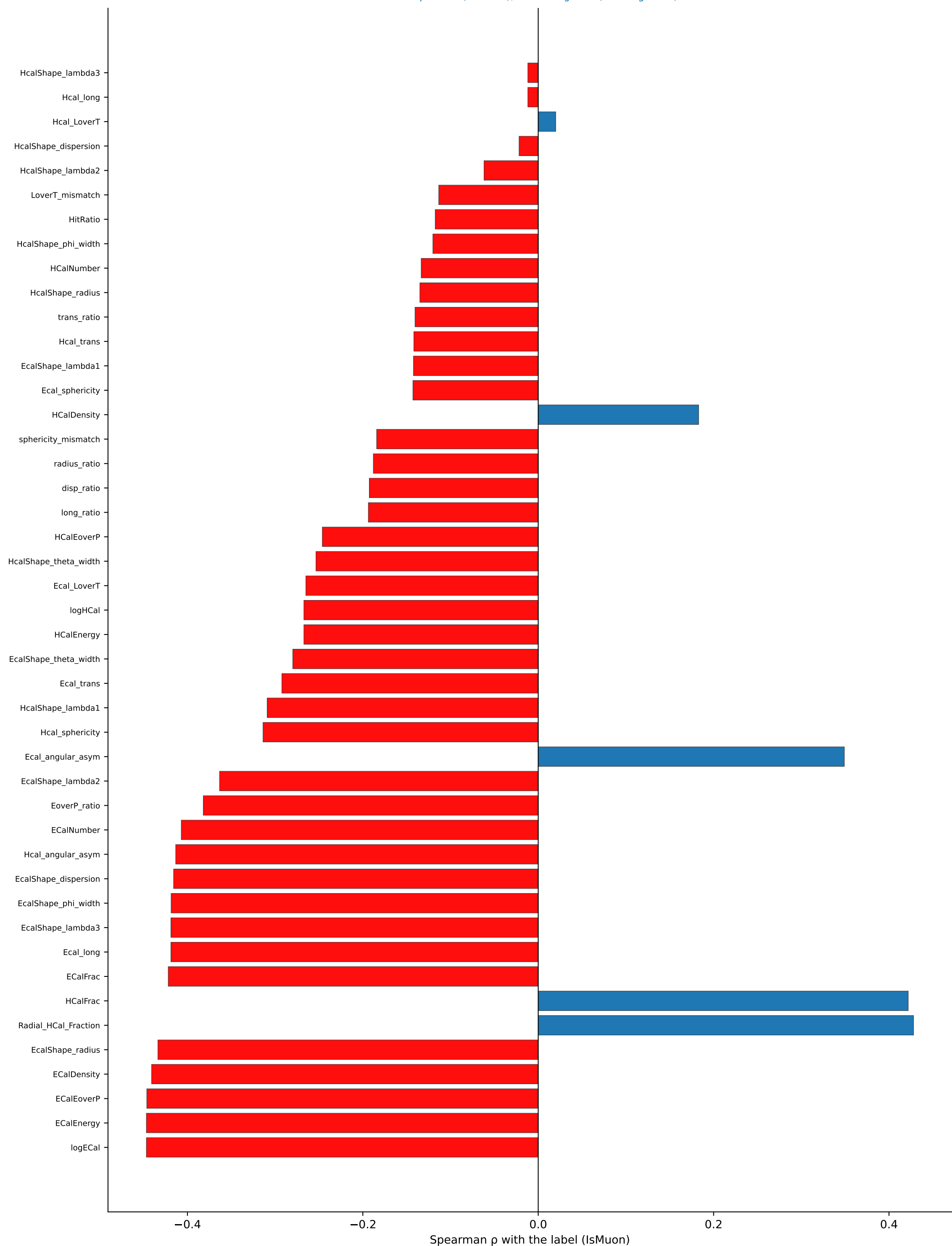


Spearman Correlation Matrix — All Features
(values from the full dataset)

Higher $|\rho|$ means a stronger monotonic relationship between features



Feature correlation with signal/background class
Blue = positive ($\rightarrow$ muon), Red = negative ($\rightarrow$ background)

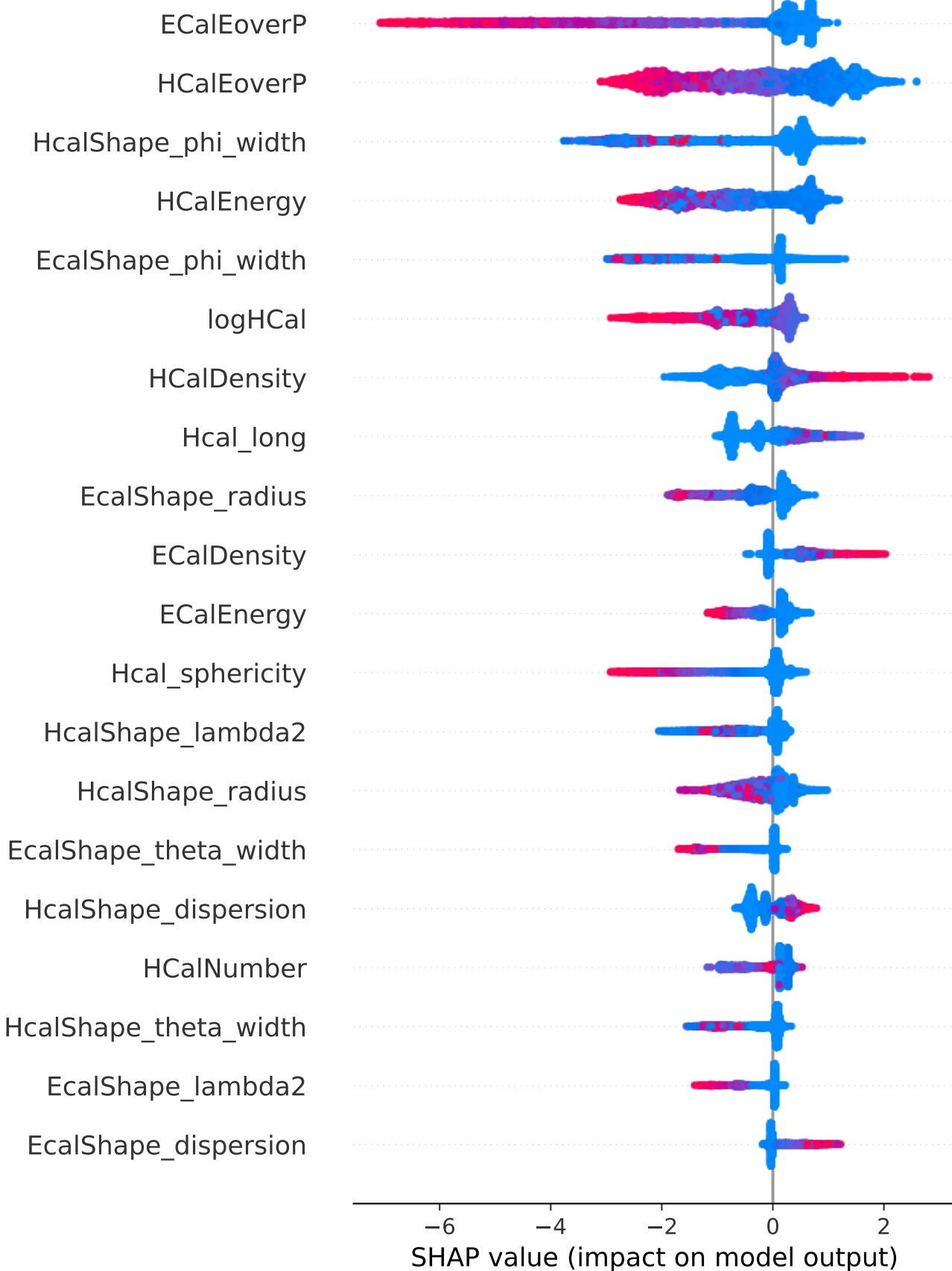


SHAP Summary Plot — Feature Impact on P(Muon)

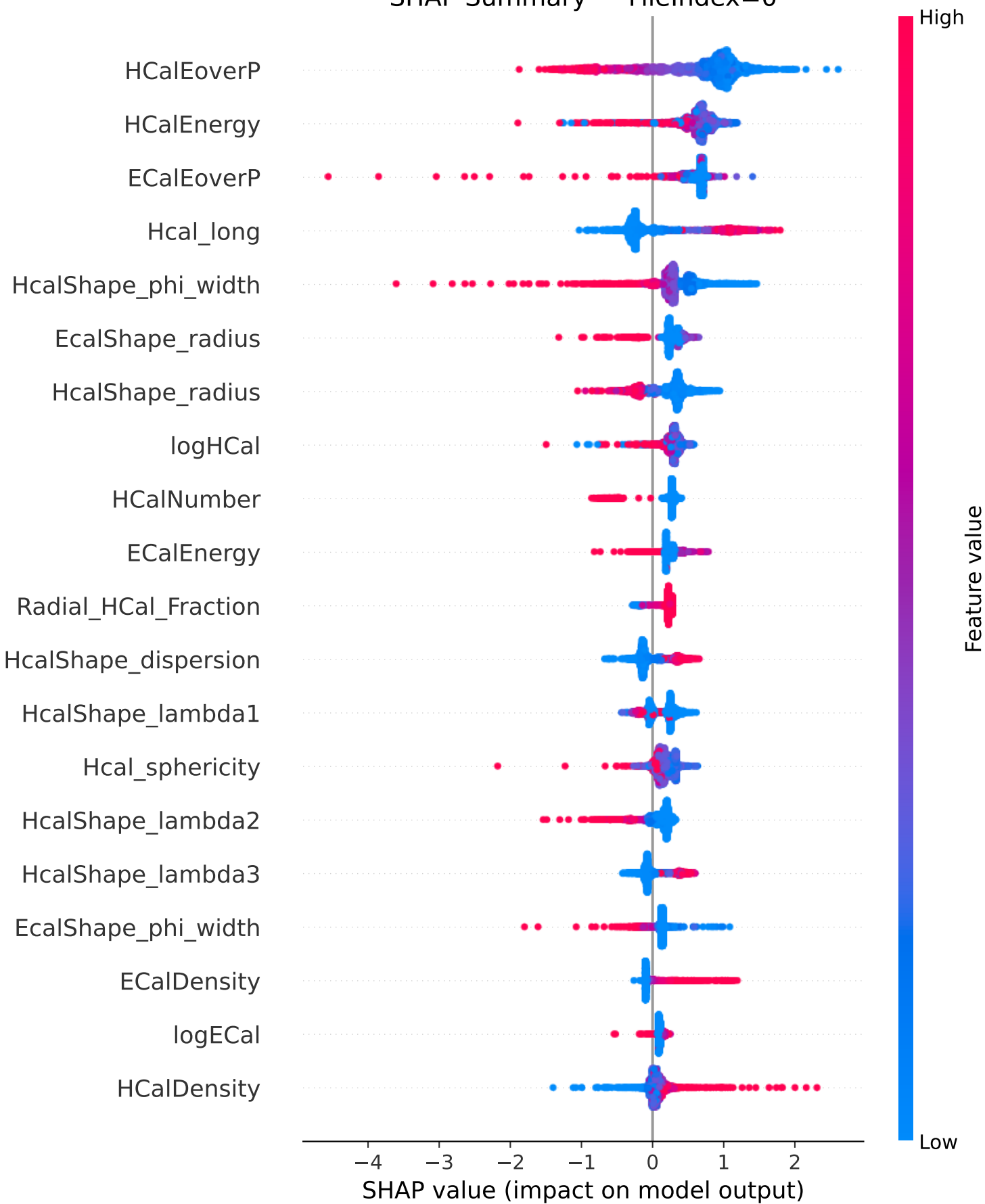
High

Feature value

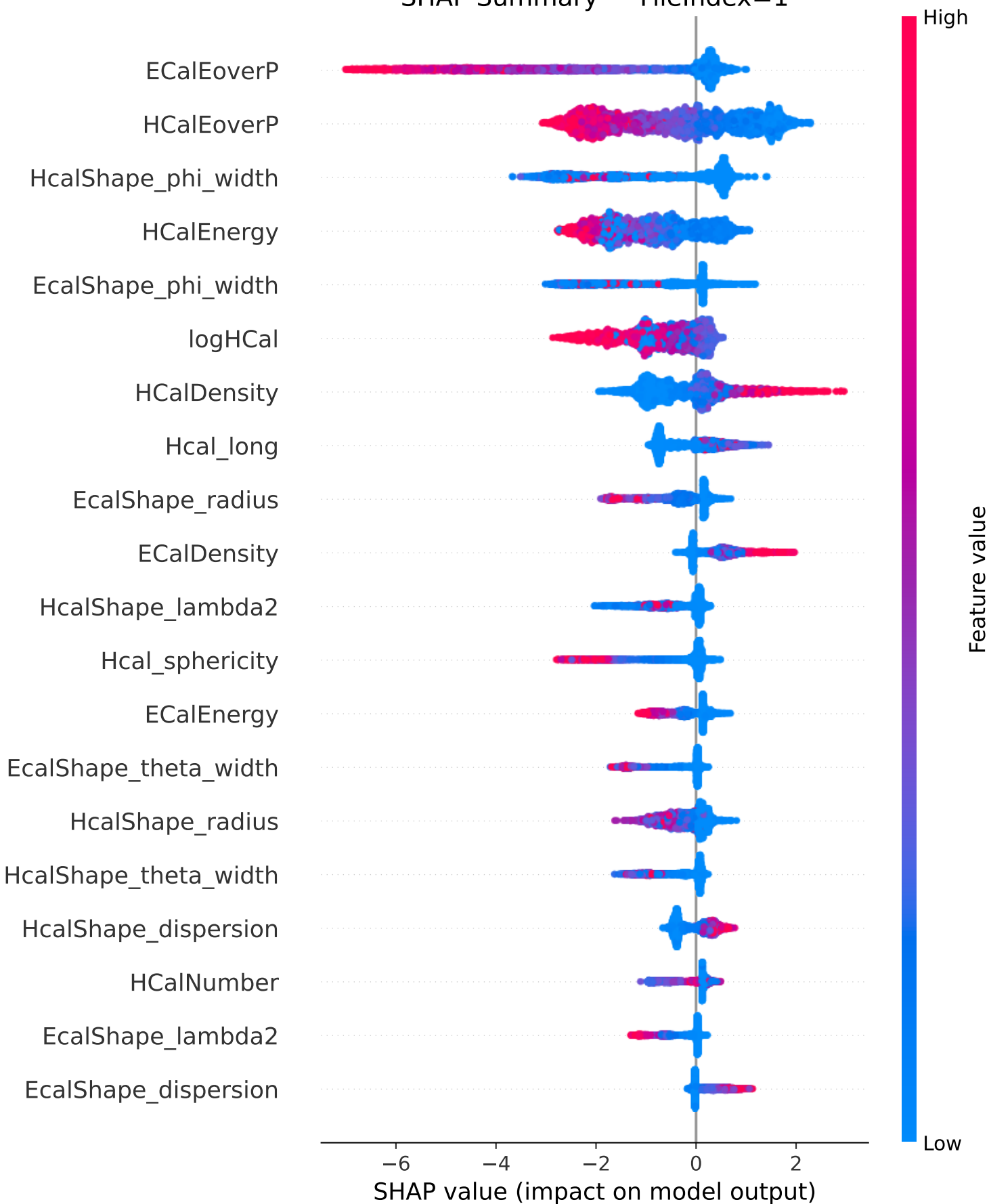
Low



SHAP Summary — FileIndex=0



SHAP Summary — FileIndex=1



SHAP Similarity Matrix — FileIndex vs FileIndex
(correlation between SHAP importance vectors)

